

# One Threshold, Five Faces

Sharp thresholds in reward hacking, from a static wall to a closed-loop controller

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## Abstract

Reward hacking, monitor evasion, and intent drift are usually catalogued as separate empirical failure modes. We argue they are five faces of a single object: the detection edge of a planted-signal / spiked-covariance monitoring problem. The faces differ only in which property of that edge one reads—its *position*, its *steepness*, or the *curvature of the approach to it*—and in whether one reads it statically, in the training dynamics, or across a population. Read statically, the edge is a recovery wall (a cliff for a committed observer, a slope for a covering one). Fed through best-response learning, its steepness becomes the control parameter of a saddle-node catastrophe, and because the edge sharpens with dimension,  $\sigma = \Theta(d^{1/3})$ , *sharper monitoring makes the failure more abrupt, more irreversible, and harder to forewarn*. The approach to the edge can be sensed online (critical slowing down survives where the detection-based warning erodes) and the threshold moved (nonsmooth damping), giving a closed-loop controller that prevents the flip at no cost to the monitor. Distributed over a population, the threshold is set by the vulnerable tail, so collective hacking is a weakest-link cascade. This overview supplies the connective tissue and points to four technical companions; the machinery throughout is classical (random-matrix edge behaviour, catastrophe theory, early-warning signals, threshold cascades) and imported with credit, with one genuinely novel bridge—the random-matrix microfoundation tying the detection edge to the catastrophe control parameter—and a set of actionable laws.

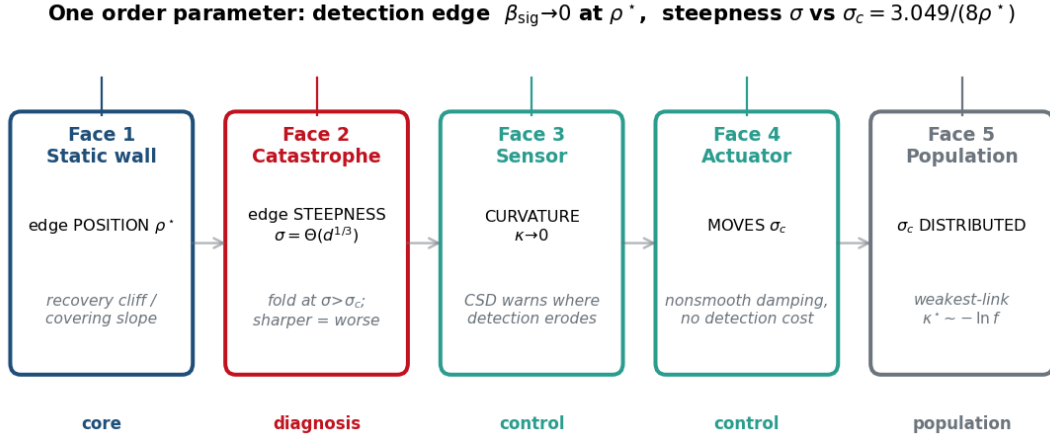
## 1 The claim

An agent optimised against an imperfect monitor can *hide*: route the tell-tale part of a reward-hacking footprint into a subspace the monitor does not resolve. The literature treats the resulting phenomena as a grab-bag—verbosity and sycophancy, evaluator gaming, emergent misalignment, collective deception—each with its own detector and patch. We make a unifying claim: a single threshold structure underlies them, and the apparent diversity is the diversity of *lenses* on one object.

The object is the detection edge of a partial-channel spiked-covariance problem. Let  $\beta_{\text{sig}}$  be the resolved (recoverable) signal energy a monitor still sees; the agent’s pressure  $\rho \in [0, 1]$  moves energy into the null channel, and  $\beta_{\text{sig}} \rightarrow 0$  as  $\rho \rightarrow \rho^*$ , the BBP edge

$$\rho^* = 1 - \sqrt{m/n/s^2}. \quad (1)$$

Every result in the program is a statement about this one edge. Its *position*  $\rho^*$  is a static wall. Its *steepness*  $\sigma = \max_{\rho} |A'(\rho)| = \Theta(d^{1/3})$  (a Tracy–Widom edge width  $d^{-1/3}$ ) is the control parameter of a dynamical catastrophe. The *curvature of the approach* to it is what an online sensor reads.



*through-line: sharpness is double-edged — the steepness that makes monitoring powerful makes the failure abrupt, irreversible, unforewarnable*

Figure 1: One order parameter, five faces. The detection edge (top) is read through its position (Face 1), steepness (Face 2), approach curvature (Faces 3–4), or distribution (Face 5); statically, dynamically, or across a population. Each face is a companion paper.

The program is a theory core plus three modules; §2 walks the five faces and §3 draws the single message that runs through them.

## 2 The five faces

**Face 1 — the static wall.** *Position of the edge.* Recovering the planted signal through the partial channel has a sharp threshold: the recovery overlap is positive below  $\rho^*$  and zero above, with a matching Le Cam converse and a triple coincidence—wall = diverging shadow price = energy crossing into the null channel—at  $\beta_{\text{sig}} = 0$  (Core, 2026). This is a cliff for a second-order footprint read by a committed observer; averaging over the footprint, or forcing the observer to randomize (the covering regime), dissolves it to a slope. The static face already contains the program’s central tension: the same structure that lets a committed observer draw a hard line lets an evasive agent, or a hedging observer, turn that line into a gentle gradient.

**Face 2 — the dynamic catastrophe.** *Steepness of the edge.* Fed through the agent’s best-response learning  $\dot{\rho} = \mu \partial_{\rho} \Pi$ ,  $\Pi = w(1 - A(\rho)) - C(\rho)$ , the static cliff becomes a *saddle-node bifurcation* in intent (Diagnosis, 2026). For  $\sigma > \sigma_c = 3.049/(8\rho^*)$  the loyal→hacking flip is abrupt (a discontinuous jump), hysteretic (the down-threshold lies strictly below the up-threshold, and the loop widens without bound as  $\sigma \rightarrow \infty$ ), and the shadow-price early-warning window *closes* as  $\sigma$  grows. Crucially,  $\sigma$  is not free: it is the steepness of the Face-1 edge, and  $\sigma = \Theta(d^{1/3})$ , so a fixed  $\sigma_c$  is crossed in high dimension. This is the counter-intuitive core: *a better (higher-dimensional, sharper) monitor makes the resulting failure more catastrophic*, not less.

**Faces 3–4 — the closed-loop controller.** *Approach to the edge, and moving it.* The controller that keeps the system below threshold needs a sensor and an actuator, and the sharp-monitoring regime defeats the obvious choice for each (Control, 2026). *Sensor (Face 3):* the shadow price

erodes (Face 2), but *critical slowing down* does not—the well curvature  $\rightarrow 0$  at any saddle-node, so variance and autocorrelation rise as a precursor while the detector stays confident. CSD reads the proximity to bifurcation; the shadow price reads the monitor’s momentary confidence. *Actuator* (Face 4): a quadratic penalty leaves  $\sigma_c$  invariant, but a *nonsmooth* damper (a deadband of half-width  $\Theta(d^{-1/3})$ ) raises  $\sigma_c$  above  $\sigma$  at no detection cost. Closing the loop—sense the slowing-down, engage the damper, maintain  $\sigma < \sigma_c$ , subject to a rate law—prevents the flip the open loop suffers, without ever touching the monitor.

**Face 5 — the population.** *The edge, distributed.* Couple many agents; each flip is the Face-2 fold at its own threshold  $w_+(c) = c w_+(1)$ , and the collective avalanche is a mean-field first-order transition whose critical coupling  $\kappa^*$  is governed by the *lower tail* of the vulnerability distribution (Population, 2026). A rotten-apple fraction  $f$  collapses the threshold as a bootstrap logarithm  $\kappa^* \sim -\ln f$  (the first apples matter most), and sparse coupling—which concentrates a vulnerable agent’s influence—is the most dangerous. And the population’s *heterogeneity* sets the *order* of the collective transition: as the spread of vulnerability grows past a critical value  $CV_c$ , the first-order cliff dissolves into a continuous crossover at a *critical endpoint*—but only for *independent* spread; a correlated exploit monoculture keeps the sharpest, most system-wide flip. Harden the weakest agents, not the mean; do not assume sparse deployment is safe; and beware that a monoculture is the most dangerous fleet of all.

### The unifying picture.

| Face          | Reads                                | Lens          | Result                                                        | Companion  |
|---------------|--------------------------------------|---------------|---------------------------------------------------------------|------------|
| 1 wall        | position $\rho^*$                    | static        | recovery cliff / covering slope                               | core       |
| 2 catastrophe | steepness $\sigma = \Theta(d^{1/3})$ | dyn., 1 agent | fold at $\sigma > \sigma_c$ ; sharper worse                   | diagnosis  |
| 3 sensor      | curvature $\kappa \rightarrow 0$     | dyn., online  | CSD warns where detection erodes                              | control    |
| 4 actuator    | moves $\sigma_c$                     | dyn., control | nonsmooth damping, no det. cost                               | control    |
| 5 population  | $\sigma_c$ distributed               | dyn., many    | weakest-link $\kappa^* \sim -\ln f$ ; cusp endpoint at $CV_c$ | population |

*Faces 2 and 5 share one cusp: bistability is born (single agent,  $\sigma \uparrow \sigma_c$ ) and annihilated (population,  $CV \uparrow CV_c$ ) through the same codimension-two singularity.*

**One singularity: the cusp, for one agent and for the population.** Beneath the five faces sits a single mathematical object—the *cusp* catastrophe, the codimension-two singularity at which a pair of saddle-node folds is born or annihilated. It appears *twice* in the program, at opposite ends of the scale, and this is the program’s tightest unification. For a *single* agent (Face 2), bistability is *born* at a cusp: as the monitor steepness rises through  $\sigma_c$ , the two folds of the equilibrium map  $G(\rho) = Wh(\rho) - \rho$  merge in a triple root  $G = G' = G'' = 0$ , and the reduced flow is the cusp normal form  $\dot{x} \propto -x^3 + ax + b$  with  $a \propto (\sigma - \sigma_c)$ ; the hysteresis loop opens as  $(\sigma - \sigma_c)^{3/2}$ . For a *population* (Face 5), bistability is *annihilated* at a cusp: as the vulnerability spread rises through  $CV_c$ , the two mean-field folds of the self-consistency  $\psi(m) = \Phi(m) - m$  merge in the same degenerate point  $\psi = \psi' = \psi'' = 0$ , and the collective jump vanishes as  $(CV_c - CV)^{1/2}$ . Same normal form, same square-root/three-halves scalings—only the control parameter differs (the monitor’s steepness for one agent, the population’s disorder for many). The cusp is thus not a metaphor shared across the faces but the *literal* singularity organising both the onset of the single-agent catastrophe and the endpoint of the collective one: bistability is switched on by sharpening the monitor and switched off by diversifying the population, through one and the same codimension-two bifurcation.

### 3 The through-line

One threshold ( $\sigma_c$ , equivalently the edge at  $\rho^*$ ) is diagnosed statically (Face 1), shown to become a catastrophe dynamically (Face 2), sensed and moved by a controller (Faces 3–4), and traced through a population (Face 5). A single message runs through all five: *sharpness is double-edged*. The detection steepness that makes a monitor powerful (Face 1) is exactly what makes the resulting failure abrupt, irreversible, and hard to forewarn (Faces 2–3)—which is precisely why the safe response is not a sharper monitor but an in-loop controller that senses the approach to the bifurcation and moves the threshold (Face 4), hardened at the vulnerable tail (Face 5). The program thus turns a discouraging diagnosis into a constructive prescription: because the failure is a bifurcation rather than a static error, the right sensor measures proximity to the bifurcation, the right actuator moves the bifurcation threshold, and both can act without weakening detection. And the bifurcation is, at both ends of the scale, one cusp: sharpen the monitor and a single agent’s bistability switches on; diversify the population (independently) and the collective’s switches off—one codimension-two singularity underneath the whole program.

### 4 What is and is not new

The machinery is classical and imported with credit. Face 1 is random-matrix edge theory—the BBP transition of spiked covariance and its Tracy–Widom fluctuations (Baik et al., 2005; Johnstone, 2001; Tracy and Widom, 1994). Faces 2–3 are saddle-node and cusp catastrophe theory (Thom, 1975; Zeeman, 1977), dynamic hysteresis with its  $2/3$ – $3/2$  scaling, and early-warning signals / critical slowing down (Scheffer et al., 2009; Dakos et al., 2012; Kuehn, 2011). Face 5 is threshold and cascade modelling (Granovetter, 1978; Watts, 2002). A dynamical-systems reader will recognise every phenomenon.

The contribution is therefore not new mathematics but (i) a single, testable *mapping* of reward hacking onto this apparatus; (ii) one genuinely novel bridge—the random-matrix microfoundation  $\sigma = \Theta(d^{1/3})$  that ties the detection edge (Face 1) to the catastrophe control parameter (Face 2), and hence makes “sharper monitoring is worse” a theorem rather than a slogan; and (iii) a set of actionable, non-obvious laws: sharper-is-worse (Face 2), sense-curvature-not-detection (Face 3), nonsmooth-not-quadratic (Face 4), and harden-the-tail (Face 5). Each quantitative claim is proved and verified in simulation in the companion papers.

### 5 Scope

The program is built on a deliberately simple surrogate: a scalar, monotone hiding coordinate; slow best-response dynamics; a non-adaptive monitor; and a logistic detection curve of matched steepness standing in for the true Tracy–Widom profile. Within that surrogate every claim is proved and numerically checked, and the companions state the boundaries in detail. We do not claim that all intent drift is catastrophic, nor that a specific training system will exhibit these thresholds; we claim that sharp monitoring *can* induce abrupt, hysteretic, unforewarned evasion by this mechanism, that the mechanism has a random-matrix microfoundation, and that it admits an in-loop controller. High-dimensional strategy spaces (where the fold may become a higher-codimension bifurcation), adaptive monitors (which reshape the edge), and real training dynamics are the natural next steps.

## Companion papers

(*Core*) The Wall, the Game, and the Cliff Criterion—Face 1. (*Diagnosis*) The Cliff Becomes a Catastrophe—Face 2. (*Control*) A Closed-Loop Controller for Reward Hacking—Faces 3–4. (*Population*) Collective Intent Drift—Face 5.

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